

Count vectorizer → Numerical Form ✓

TF-IDF → Term Frequency - Inverse Document Frequency.

$$\text{idf}(t) = \log \frac{1+N}{1+\text{df}(t)} + 1$$

text	sentiment
"movie is good"	positive
"movie is bad"	negative

movie good bad

vocabulary → ["bad", "good", "movie"] corpus

doc	bad	good	movie
<u>d_1 (movie is good)</u>	0	1	1
<u>d_2 (movie is bad)</u>	1	0	1

$\text{df} \rightarrow (\text{bad}) \rightarrow 1$ $(\text{good}) \rightarrow 1$ $\text{movie} \rightarrow (2)$ $(N=2)$

TF-IDF

$$\begin{aligned} \text{idf}(\text{good}) &= \log \left(\frac{1+N}{1+\text{df}(t)} \right) + 1 = \log \left(\frac{1+2}{1+1} \right) + 1 \\ &= \log \left(\frac{3}{2} \right) + 1 \\ &= \log 1.5 + 1 \end{aligned}$$

$$= \frac{100(\frac{1}{2})}{100} + 1$$

$$= \log(1.5) + 1$$

$$= 0.4054 + 1 = 1.4054$$

$$\text{idf}(\text{bad}) = \log\left(\frac{1+N}{\text{idf}(t)}\right) + 1 = \log\left(\frac{1+2}{1+1}\right) + 1$$

$$= \log 3/2 + 1$$

$$= 1.4045$$

$$\text{idf}(\text{"movie"}) = \log\left(\frac{1+N}{\text{idf}(t)}\right) + 1$$

$$= \log\left(\frac{1+2}{1+2}\right) + 1$$

$$= \log \frac{3}{2} + 1$$

$$= \log 1 + 1 = 0 + 1 = \underline{\underline{1}}$$

TF-IDF

$d_0(\text{movie in good})$

$d_1(\text{movie in bad})$

~~good~~ good movie

0 1.4054 1

1.4054 0 1

✶ unit normalization = $\frac{x}{\sqrt{\sum x_i^2}}$

$d_0 \rightarrow [0, 1.4054, 1]$

$$= \sqrt{0^2 + (1.4054)^2 + 1^2} = \sqrt{1^2 + (1.4054)^2}$$

$$= 1.724$$

$$\frac{0}{1.724} \quad \frac{1.4054}{1.724} \quad \frac{1}{1.724}$$

$$\underline{d_0} \quad \underline{[0 \quad 0.8148 \quad 0.5797]} \rightarrow \textcircled{1}$$

$$\underline{d_1} \quad [0.5797 \quad 0.8148 \quad 0.5797]$$

$$\underline{X_{\text{test}}}$$
 countVect. fit - train(df.test)

$$\underline{x_{\text{train}}}, \underline{m_{\text{test}}}, \underline{y_{\text{test}}}, y_{\text{test}} = \text{train-test-split}(x, y)$$

$$\underline{y_{\text{test}}}$$